

Hypothesis:

If we say that Simple Total increments the total exactly $1 + \sum_{k=5n}^n k$ times for some input of n , we can conclude the Big-O time complexity is $O(n^2)$ if, for a given input n and a positive constant C

$$1 + \sum_{k=5n}^n k \leq C \cdot n^2$$

The summation can be written as

$$1 + \sum_{k=5n}^n k = 1 + \sum_{k=1}^n k - \sum_{k=1}^{5n} k = 1 + \frac{n(n+1)}{2} - \frac{5n(5n+1)}{2}$$

$$\Rightarrow 12n^2 + 6n + 1$$

Similarly, we can write the following inequality substituting the summation with our polynomial

$$12n^2 + 6n + 1 \leq C \cdot n^2$$

In order to prove this for a growing input n , we first need a base case. We choose $n_0 = 1$ as a positive non-zero input.

$$12(1) + 6(1) + 1 \leq C \cdot (1)^2$$
$$19 \leq C$$

The inequality is true where $C \geq 19$. Now we show that this relation holds for a growing input, say $n = x + 1$

$$12(x+1)^2 + 6(x+1) + 1 \leq C \cdot (x+1)^2$$

$$12(x^2 + 2x + 1) + 6x + 7 \leq C(x^2 + 2x + 1)$$

$$12x^2 + 24x + 12 + 6x + 7 \leq C(x^2 + 2x + 1)$$

$$12x^2 + 30x + 19 \leq C(x^2 + 2x + 1)$$

Where $C = 19$, we prove that SimpleTotal is bound
by $O(n^2)$.

$$12x^2 \leq 19x^2$$